and at the same time, vent outlets should direct the vapour discharge upwards in the form of unimpeded jets.
RESOLUTION MEPC.55(33) (adopted on 30 October 1992)
ADOPTION OF AMENDMENTS TO THE INTERNATIONAL CODE FOR THE CONSTRUCTION AND EQUIPMENT OF SHIPS CARRYING
DANGEROUS CHEMICALS IN BULK (IBC CODE)- ; -
8.2.2 The venting systems should be connected t o the top of each cargo tank snd as far as practicable the cargo vent lines should be self-draining back to the cargo tanks under all normal operational conditions of li st and trim. Where it is necessary to drain venting systems above the l evel of sny

---CHUNK---

SOURCE: MEPC.55(33).pdf
snd as far as practicable the cargo vent lines should be self-draining back to the cargo tanks under all normal operational conditions of li st and trim. Where it is necessary to drain venting systems above the l evel of sny
pressure/vacuum valve, capped or plugged drain cocks should be provided.
8.2.3 Provision should be made to ensure that the liquid head in any tank does not exceed the design head of the tank. Suitable high-level alarms, overflow control systems or spill valves, together with gauging snd tank filling procedures may be accepted for this purpose. Where the means of limiting cargo tank overpressure includes an automatic closing valve, the valve should comply with the appropriate provisions of 15.19.
8.2.4 Tank venting systems should be designed and operated so as to ensure that neither pressure nor vacuum created in the cargo tanks during loading or unloading exceeds ta nk design parameters. The main factors to be considered in the sizing of a tank venting system are as fo l lows:
.1 design loading and unloading rate;
.2 gas evolution during loading: this should be taken account of by

---CHUNK---

SOURCE: MEPC.55(33).pdf
unloading exceeds ta nk design parameters. The main factors to be considered in the sizing of a tank venting system are as fo l lows:
.1 design loading and unloading rate;
.2 gas evolution during loading: this should be taken account of by multiplyi ng the maximum loading rate by a factor of at least 1.25;
.3 density of the cargo vapour mixture;
.4 pressure loss in vent piping and across valves and fittings;
.5 pressure/vacuum settings of relief devices.
8.2.5 Tank vent piping connected to cargo tanks of corrosion resistant material, or to tanks which are lined or coated to hand l e special cargoes as required by the Code, should be similarly lined or coated or constructed of corrosion-resistant material.
8.2.6 The master should be provided with the maximum permissible loading and unloading rates for each tank or group of tanks consistent with design of the venting systems.
8.3 Types of tank venting systems
8.3.1 An open tank venting system is a system which offers no restriction except for friction losses to the free flow of cargo vapours to and from the carg o tanks during normal operations. An open venting system may consist of

---CHUNK---

SOURCE: MEPC.55(33).pdf
8.3.1 An open tank venting system is a system which offers no restriction except for friction losses to the free flow of cargo vapours to and from the carg o tanks during normal operations. An open venting system may consist of individual vents from each tank, or such individual vents may be combined into a common header or headers, with due regard to cargo segregation. In no case should shutoff valves be fitted either to the individual vents or to the header.
8.3.2 A controlled tank venting system is a system in which pressure and vacuum relief valves or pressure/vacuum valves are fitted to each tank to limit the pressure or vacuum in the tank. A controlled venting system may consist of individual vents from each tank or such individual vents on the pressure side only as may be combined into a common header or headers with due regard to cargo segregation. In no case should shutoff valves be fitted
RESOLUTION MEPC.55(33) (adopted on 30 October 1992)
ADOPTION OF AMENDMENTS TO THE INTERNATIONAL CODE FOR THE CONSTRUCTION AND EQUIPMENT OF SHIPS CARRYING
DANGEROUS CHEMICALS IN BULK (IBC CODE)- 5 -

---CHUNK---

SOURCE: MEPC.55(33).pdf
RESOLUTION MEPC.55(33) (adopted on 30 October 1992)
ADOPTION OF AMENDMENTS TO THE INTERNATIONAL CODE FOR THE CONSTRUCTION AND EQUIPMENT OF SHIPS CARRYING
DANGEROUS CHEMICALS IN BULK (IBC CODE)- 5 -
either ab ove or below pressure or vacuum relief valves or pressure/vacuum valves. Provision may be made for bypassing a pressure or vacuum valve or pressure/vacu um valve under ce rtain operating co nditi ons provided that the requirement of 8.3 .5 is maintained and that there is suitable indication to show whether or not the valve is bypassed.
8.3.3 The position of vent outlets of a controlled tank venting sys tem should be arranged:
.1 at a height of not less thsn 6 m above the weather deck or above a raised walkway if fitted within 4 m of the raised walkway;
.2 at a distance of at least 10 m measured horizontally from the neare s t air intake or opening to accommodation, service and machi nery spaces and ignition sources.
8.3.4 The vent outlet height re fer red to in 8.3.3.1 may be reduced to 3 m above the deck or a raised walkway, as applicable, provided that hi gh velocity venting valves of a type approved by the Administration directing the

---CHUNK---

SOURCE: MEPC.55(33).pdf
8.3.4 The vent outlet height re fer red to in 8.3.3.1 may be reduced to 3 m above the deck or a raised walkway, as applicable, provided that hi gh velocity venting valves of a type approved by the Administration directing the vapour/air mixture upwards in an unimpeded jet with an exit velocity of at least 30 m/s are fi t ted.
8.3.5 Controlled tank venting systems fitted to tanks to be used for cargoes having a f lashpoint not exceeding 60·C (closed cu p test) should be provided

with devices to prevent the passage of flame into the cargo tanks. The design, testing and locating of the devices should comply with the requirements of the Administration which should contain at least the standards adopted by the Organization.*

8.3.6 In designing ve nti ng sy st ems and i n the selection of devices to prevent the passage of flsme for incorporation into the tank vent i ng system, due attention should be paid to the possibility of the blockage of these systems and f ittings by, for example, the freezing of cargo vapour, polymer build up, atmospheric dust or icing up in adverse weather conditions. In this context

---CHUNK---

SOURCE: MEPC.55(33).pdf

attention should be paid to the possibility of the blockage of these systems and f ittings by, for example, the freezing of cargo vapour, polymer build up, atmospheric dust or icing up in adverse weather conditions. In this context it should be noted that flame arresters and flame screens are more sus ceptible to blockage. Provisions should be made such that the system and fittings may be inspected, ope rat io nall y c he cked, clea ned or re newed as applica ble.

8.3.1 Reference in 8 .3.1 and 8.3.2 to the use of shutoff valves in the venting li ne s should be int erpreted to exte nd to all other means of stoppage including spectacle blanks an d blank flanges.

8.4 Ventin g requi re ments for individual pr odu c ts

Venting requirements for indi vidual products are shown in column "g" an d additional requirements in column "0" in the table of c hapt er 11.

* Ref ere nce is made to the Revised Standards for the Design, Testing and Locating of Devices to Prevent the Passage of Flame into Cargo Tanks in Tank e r s (MSC/Circ.373/Rev.l).

RESOLUTION MEPC.55(33) (adopted on 30 October 1992)

ADOPTION OF AMENDMENTS TO THE INTERNATIONAL CODE FOR THE CONSTRUCTION

---CHUNK---

SOURCE: MEPC.55(33).pdf

Locating of Devices to Prevent the Passage of Flame into Cargo Tanks in Tank e r s (MSC/Circ.373/Rev.l).

RESOLUTION MEPC.55(33) (adopted on 30 October 1992)

ADOPTION OF AMENDMENTS TO THE INTERNATIONAL CODE FOR THE CONSTRUCTION AND EQUIPMENT OF SHIPS CARRYING

DANGEROUS CHEMICALS IN BULK (IBC CODE)- 6 -

8.5 Cargo tank gas-freeing*

8.5.1 The arrangements for gas-freeing cargo tanks used for cargoes other than those for which open venting is permitted should be such as to minimize the hazards due to the dispersal of flammable or toxic vapours in the atmosphere and to flammable or toxic vapour mixtures in a cargo tank. Accordingly, gas-freeing operations should be carried out such that vapour is initially discharged:

.1 through the vent outlets specified in 8.3 . 3 and 8.3.4; or

.2 through outlets at least 2 m above the cargo tank deck level with a vertical efflux velocity of at l east 30 mls maintained during the gas-freeing operation; or

.3 through outlets at l east 2 m above the cargo tank deck leve l with a vertica l efflux velocity of at least 20 mls which are protected by suitable devices to prevent the passage of flame.

---CHUNK---

SOURCE: MEPC.55(33).pdf
gas-freeing operation; or
.3 through outlets at l east 2 m above the cargo tank deck leve l with a vertica l efflux velocity of at least 20 mls which are protected by suitable devices to prevent the passage of flame.
When the flammable vapour concentration at the outlets has been reduced to 30% of the lower flammable limit and in the case of a toxic product the vapour concentration does not present a significant health hazard, gas-freeing may thereafter be continued at cargo tank deck level.
8.5.2 The outlets referred to in 8.5.1.2 and 8.5.1.3 may be fixed or portable pipes.
8.5.3 In designing a gas-freeing system in conformity with 8.5.1 partic ularly in order to achieve the required exit velocities of 8.5.1.2 and 8.5.1.3, due consideration should be given to the following:
.1 materials of construction of system;
.2 time to gas-free;
.3 f l ow characteristics of fans to be used;
.4 the pressure losses created by ducting, piping, cargo tank inlets and outlets;
* Reference is made to the Revised Factors to be taken into Consideration